

TRACK SHIELD:

AN INTELLIGENT AI-IOT SYSTEM FOR RAILWAY TRACK HEALTH AND INTRUSION DETECTION

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BONAFIDE CERTIFICATE

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ABSTRACT

Railway transportation requires continuous monitoring to ensure the safety of passengers, railway infrastructure, And surrounding areas. Track damage, abnormal vibrations, and unauthorized intrusion onto railway tracks can Lead to serious accidents and operational disruptions. This project proposes TrackShield, and intelligent AI-IoTBased railway safety monitoring system that combines track condition monitoring with real-time intrusion Detection. The system uses an ESP32 microcontroller to collect data from sensors such as an accelerometer and Vibration sensor for identifying abnormal track conditions. A HuskyLens AI camera is integrated to detect and Recognize people or objects entering the restricted railway track area. When abnormal conditions or Unauthorized intrusion are detected, the system generates an immediate warning through a buzzer and display And can transmit alerts through an IoT platform. The proposed system is designed as a low-cost prototype that Demonstrates continuous railway safety monitoring and early warning capabilities. By integrating sensor-based Monitoring, AI-based visual detection, and Railway transportation requires continuous monitoring to ensure the Safety of passengers, railway infrastructure, and surrounding areas. Track damage, abnormal vibrations, and Unauthorized intrusion onto railway tracks can lead to serious accidents and operational disruptions. This project Proposes TrackShield, an intelligent AI-IoT based railway safety monitoring system that combines track Condition monitoring with real-time intrusion detection. The system uses an ESP32 microcontroller to collect Data from sensors such as an accelerometer and IoT communication, TrackShield aims to provide a reliable and Efficient approach to improving railway safety and reducing the risk of accidents caused by track abnormalities And unauthorized access.

KEYWORD: Railway Safety; HuskyLens; Intrusion Detection; IoT; Track Monitoring.

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CHAPTER 1.INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Introduction

Railway transportation is one of the most important modes of transportation, connecting people and goods across different locations. Railway track safety is essential for ensuring safe and reliable train operations. Track damage, unusual vibrations, and unauthorized intrusions near railway tracks may create safety risks. Therefore, continuous monitoring and early identification of abnormal conditions are important for improving railway safety.

The Track Shield project proposes an Intelligent railway track monitoring system that combines Artificial Intelligence (AI), the Internet of Things (IoT), and sensor technology. The system uses an ESP32 microcontroller connected to a vibration sensor to identify unusual vibrations or disturbances. An LCD displays the system status, and a buzzer provides an audible alert when abnormal vibration is detected.

Problem Statement

Traditional railway track inspection often depends on manual checking and periodic maintenance. These methods may not identify every abnormal condition immediately. The major problems include:

- **Delayed Detection:** Track abnormalities may remain unnoticed between inspections.
- **Manual Monitoring:** Regular inspection requires time and human effort.
- **Limited Real-Time Alerts:** Conventional inspection methods may not provide immediate warnings when disturbances occur.
- **Safety Risks:** Undetected track problems or unauthorized disturbances near the railway track may increase safety risks.
- **Objectives of the Project**

The main objectives of Track Shield are:

- To develop an IoT-based railway track monitoring prototype using an ESP32 microcontroller.
- To detect unusual vibrations using a vibration sensor.
- To provide audible alerts through a buzzer when abnormal vibration is detected.

- To display the monitoring status using an LCD.
- To explore remote monitoring through IoT connectivity.
- To support faster identification of abnormal conditions and improve railway safety awareness.

Scope of the Project

The project focuses on developing a small-scale railway track monitoring prototype. Its main functions include vibration sensing, processing sensor signals through the ESP32, displaying system status, and activating an alert when the configured detection condition is met.

The prototype demonstrates the basic concept of railway track monitoring using affordable electronic components. Additional sensors, communication modules, and AI-based analysis can be incorporated in future versions to improve detection capabilities. The current vibration-based prototype alone cannot confirm actual rail cracks or identify every type of intrusion.

System Overview

The Track Shield prototype consists of the following main components:

1. **ESP32 Microcontroller:** Processes the sensor signals and controls the output devices.

2. **Vibration Sensor:** Detects vibrations or disturbances in the monitored setup.

3. **LCD Display:** Displays the system status and relevant messages.

4. **Buzzer:** Produces an audible warning when the programmed abnormal-vibration condition is detected.

5. **IoT Connectivity:** The ESP32 can support data transmission to a compatible platform when remote monitoring is implemented.

Advantages of the System

- Provides a simple method for monitoring unusual vibrations.
- Generates an audible alert when the configured detection condition is met.
- Displays the system status clearly.
- Uses relatively affordable and widely available electronic components.
- Provides a foundation for future IoT-based railway monitoring systems.

Applications of the Project

The Track Shield concept may be developed for railway safety demonstrations, educational prototypes, experimental track monitoring, and future sensor-based railway inspection systems. With further development and validation, additional sensors and communication technologies could support more comprehensive railway monitoring.

Expected Outcomes

The expected outcome is a working prototype that uses an ESP32 and vibration sensor to detect disturbances, displays the status on an LCD, and activates a buzzer when the programmed condition is satisfied. The project demonstrates how embedded systems and IoT technologies can contribute to railway monitoring and provide a foundation for more advanced safety applications.

1.2 SCOPE AND OBJECTIVE

Scope of the Project

The scope of the Track Shield project is to develop a prototype for railway track health monitoring and disturbance detection using sensor technology and an ESP32 microcontroller. The system focuses on identifying unusual vibrations and generating alerts to support railway safety monitoring.

The current prototype is intended for educational and experimental purposes. It demonstrates the basic concept of sensor-based railway monitoring and can be expanded in the future by integrating additional sensors, advanced AI techniques, and remote monitoring facilities.

The scope includes:

- Monitoring vibration and disturbances in a small-scale railway track model.
- Processing sensor signals using the ESP32 microcontroller.
- Displaying the monitoring status on an LCD.
- Activating a buzzer when the configured abnormal-vibration condition is detected.
- Providing a foundation for future improvements in railway track safety monitoring.

Objectives of the Project

The primary objective of Track Shield is to develop an intelligent monitoring prototype that detects unusual vibrations and provides timely alerts.

The specific objectives are:

1. To design a railway track monitoring prototype using an ESP32 microcontroller.

2. To detect unusual vibrations using a vibration sensor.

3. To process sensor signals and identify disturbances based on programmed conditions.

4. To activate a buzzer when abnormal vibration is detected.

5. To display system status and alerts using an LCD.

6. To explore IoT connectivity for remote monitoring of sensor information.

7. To develop a cost-effective prototype that can be enhanced with additional sensors and AI-based analysis in the future.

Expected Benefits

Track Shield aims to demonstrate how embedded systems and IoT technologies can support railway monitoring. The prototype can provide immediate local alerts for detected disturbances, reduce the need for continuous manual observation during demonstrations, and serve as a foundation for developing more advanced monitoring systems. Actual railway deployment would require extensive testing, validation, and integration with certified railway safety systems.

CHAPTER 2. LITERATURE SURVEY

2.1 LITERATURE OVERVIEW

Railway track monitoring is an important area of research that focuses on improving railway safety, reliability, and maintenance. Traditional railway inspection methods often involve manual checking and scheduled maintenance to identify defects and abnormal conditions. Although these methods are useful, they may require considerable time and may not provide continuous monitoring of railway tracks.

Recent developments in sensor technology, embedded systems, Artificial Intelligence (AI), and the Internet of Things (IoT) have created opportunities for automated railway monitoring.

Sensors can collect information about vibrations and other physical conditions, while microcontrollers process the collected signals. IoT technology can support the transmission of monitoring data to remote platforms, enabling users to observe system status without being physically present near the monitoring equipment.

The Track Shield project is developed as a small-scale railway monitoring prototype using an ESP32 microcontroller, a vibration sensor, an LCD, and a buzzer. The vibration sensor detects disturbances, the ESP32 processes the sensor signals, the LCD displays the system status, and the buzzer provides anaudible alert when the programmed condition is satisfied.

The main purpose of this literature overview is to understand the technologies used in railway monitoring and identify the need for a simple, cost-effective prototype. Track Shield demonstrates the basic concept of sensor-based disturbance detection and provides a foundation for future improvements, including additional sensors, remote data transmission, and validated AI-based analysis.

2.2 LITERATURE SURVEY

The literature survey examines existing research and technologies related to railway track monitoring, vibration detection, IoT connectivity, and AI-based fault identification. These studies provide a foundation for developing the Track Shield project.

1.Condition Monitoring of Railway Tracks from Car-Body Vibration Using a Machine Learning Technique (2019)

Hitoshi Tsunashima proposed a railway track condition monitoring method using vibration measurements collected from railway vehicles. Machine learning techniques were used to identify track irregularities from vibration data. The study demonstrated the potential of vibration analysis for railway condition monitoring. However, the approach requires suitable vibration measurements and classification methods for reliable fault identification.

2.A Machine-Learning-Based Approach for Railway Track Monitoring Using Acceleration Measured on an In-Service Train (2023)

Abdollah Malekjafarian and co-authors proposed a method for monitoring railway tracks using acceleration data measured from a train in service. The approach aimed to identify track sub-layer stiffness changes using machine learning. The study highlighted the usefulness of acceleration measurements for railway condition monitoring.

3.An Unsupervised Anomaly Detection Framework for Onboard Monitoring of Railway Track Geometrical Defects Using One-Class Support Vector Machine (2024)

This study introduced an AI-based method for identifying railway track geometry abnormalities using vibration data collected from a measuring train. A One-Class Support Vector Machine was used to detect unusual patterns associated with track conditions. The method shows the potential of anomaly detection when labelled fault data is limited. However, its performance depends on the quality of the input data and the characteristics of the monitored railway tracks.

4.Deep Learning-Based Intelligent System for Railway Track Monitoring and Fault Detection (2026)

Sujatha Krishna, Rajesh Natarajan, and Francesco Flammini proposed a railway track fault detection system based on deep learning and image classification.

The study compared different pretrained convolutional neural network models to identify defective and non-defective railway track images. Although image-based methods can recognize visible defects, their performance may be affected by image quality, environmental conditions, and the diversity of the training dataset.

Research Gap

The reviewed studies demonstrate that vibration sensing, machine learning, and image processing can support railway track condition monitoring. However, different approaches may require specialized sensors, suitable datasets, computational resources, or additional processing systems.

Track Shield addresses the need for a simple and affordable educational prototype by combining an ESP32 microcontroller, vibration sensor, LCD, and buzzer. The prototype detects unusual vibrations and provides a local alert. Future development can incorporate additional sensors, validated AI models, and IoT-based remote monitoring to improve its capabilities.

CHAPTER 3-SYSTEM DESIGN

3.1 EXISTING SYSTEM

Overview

The existing railway track monitoring system mainly relies on manual inspection, periodic maintenance, and specialized inspection equipment. Railway inspectors regularly examine tracks to identify cracks, misalignment, damaged components, and other abnormalities. Advanced monitoring systems may use cameras, ultrasonic sensors, vibration sensors, and other technologies to inspect railway track conditions.

However, traditional railway track inspection methods have several limitations. Manual inspection requires considerable time and effort, especially when examining long railway routes. The accuracy of inspection may depend on the experience of the inspectors, and problems may remain undetected between scheduled inspections. Advanced inspection equipment can also be expensive to install and maintain. In addition, systems without continuous monitoring may not provide immediate alerts when abnormal conditions occur.

The existing system therefore crea”es a need for an affordable and efficient monitoring approach that can detect unusual disturbances and provide alerts.

To address these limitations, the proposed Track Shield system uses an ESP32 microcontroller, a vibration sensor, an LCD display, and a buzzer. The vibration sensor detects disturbances, and the ESP32 processes the sensor signals. The LCD displays the system status, while the buzzer provides an alert when the programmed abnormal-vibration condition is detected.

The proposed prototype demonstrates basic railway track disturbance monitoring at a low cost. However, vibration detection alone cannot confirm actual rail cracks or identify every type of intrusion. Further development and testing with additional sensors and validated detection methods are required for reliable real-world railway applications.

3.2 PROPOSED SYSTEM

Overview

The proposed system, Track Shield, is an intelligent railway track monitoring prototype designed to detect abnormal vibrations and provide alerts. It uses an ESP32 microcontroller, a vibration sensor, an LCD display, and a buzzer to monitor disturbances and indicate abnormal conditions.

The vibration sensor detects vibrations or disturbances in the railway track model and sends signals to the ESP32 microcontroller. The ESP32 processes the sensor input and checks whether the detected vibration meets the predefined threshold condition. Based on the result, the system updates the status on the LCD display. When an abnormal vibration condition is detected, the buzzer produces an alert to notify the user.

The proposed system aims to provide a simple, low-cost, and easy-to-understand approach to railway track disturbance monitoring. It reduces the need for continuous manual observation during prototype testing and

demonstrates how sensor-based technology can support railway safety monitoring.

The main components of the proposed system are:

1. ESP32 Microcontroller: Processes sensor signals and controls the output devices.

2. Vibration Sensor: Detects vibrations and disturbances in the railway track model.

3. LCD Display: Displays the monitoring status and detected conditions.

4. Buzzer: Produces an audible alert when the programmed abnormal-vibration condition is detected.

5. Power Supply: Provides the required power to the electronic components.

The system is designed as an educational prototype. Although it demonstrates abnormal-vibration detection, the current sensor arrangement cannot independently confirm actual railway track cracks or detect all types of intrusion. Additional sensors, validated AI techniques, and extensive field testing would be required before real railway deployment. IoT-based remote monitoring can also be incorporated as a future enhancement.

3.3 SYSTEM ARCHITECTURE

The system architecture of Track Shield illustrates the arrangement and interaction of the hardware components used for railway track disturbance monitoring. The system consists of a vibration sensor, an ESP32 microcontroller, an LCD display, and a buzzer. Each component performs a specific function in detecting and indicating abnormal vibration conditions.

The vibration sensor is placed on the railway track model to detect vibrations or disturbances. The sensor sends its output signal to the ESP32 microcontroller. The ESP32 processes the

received signal and checks whether the vibration exceeds the predefined threshold. Based on the sensor input, the microcontroller controls the LCD display and buzzer.

The LCD display provides information about the system status, helping the user understand whether the detected vibration is within the programmed range or an abnormal condition has been detected. When the specified abnormal-vibration condition occurs, the ESP32 activates the buzzer to provide an audible alert.

The system architecture follows a simple input-processing-output structure. The vibration sensor acts as the input device, the ESP32 performs the processing, and the LCD display and buzzer act as output devices. This arrangement provides a basic and low-cost method for demonstrating railway track disturbance monitoring.

Main Components of the System Architecture

1. Input Unit: The vibration sensor detects disturbances in the railway track model.

2. Processing Unit: The ESP32 receives and processes the sensor signals according to the programmed logic.

3. Display Unit: The LCD displays the system status and monitoring information.

4. Alert Unit: The buzzer produces an audible alert when the programmed abnormal-vibration condition is detected.

5. Power Supply Unit: Supplies the electrical power required by the system components.

The proposed architecture can be extended in the future by integrating additional sensors, validated AI-based analysis, and IoT connectivity for remotemonitoring. The current prototype demonstrates vibration-based alerts and does not independently confirm actual railway track cracks or all types of intrusion.

3.4 HARDWARE REQUIREMENTS

The Track Shield system requires a set of hardware components to detect abnormal vibrations and provide alerts. These components work together to monitor disturbances in the railway track model and display the system status.

1.ESP32 Microcontroller

The ESP32 is the main controller of the Track Shield system. It receives signals from the vibration sensor, processes the input according to the programmed logic, and controls the LCD display and buzzer.

2.Vibration Sensor

The vibration sensor is used to detect vibrations and disturbances in the railway track model. It sends an output signal to the ESP32 whenever vibration is detected. The sensor helps identify abnormal vibration conditions based on the system's programmed settings.

3.LCD Display

The LCD display is used to show the monitoring status and messages generated by the system. It helps the user observe the system's response during testing.

4.Buzzer

The buzzer is an audible alert device. It is activated by the ESP32 when the detected vibration meets the programmed abnormal-condition criteria. This helps notify the user about a detected disturbance.

5.Breadboard and Connecting Wires

The breadboard and connecting wires are used to establish electrical connections between the ESP32, vibration sensor, LCD display, and other components. They simplify circuit assembly and testing.

6.Power Supply and USB Cable

The pow er supply provides the electrical energy required for the circuit. A USB cable can be used to power the ESP32 and connect it to a computer for uploading the program and monitoring serial output.

7.Railway Track Model

The railway track model provides a demonstration platform for testing the vibration sensor. It allows the system to be tested under controlled conditions without connecting it to an actual railway track.

These hardware components form the basic Track Shield prototype. The system demonstrates vibration-based disturbance detection and alert generation. Additional hardware, such as suitable intrusion sensors and track-condition monitoring sensors, may be required for more advanced railway safety applications.

3.5 SOFTWARE REQUIREMENTS

The Track Shield system requires software tools to develop, upload, and execute the program that controls the hardware components. The software enables the ESP32 to receive vibration sensor signals, process the input, display the system status, and activate the buzzer when an abnormal vibration condition is detected.

1.Arduino IDE

The Arduino IDE is used to write, edit, compile, and upload the program to the ESP32 microcontroller. It provides a simple development environment for programming and testing the system.

2.Embedded C/C++ Programming

Embedded C/C++ is used to develop the program for the ESP32. The program reads the vibration sensor output, checks the programmed condition, updates the LCD display, and controls the buzzer.

3.ESP32 Board Support Package

The ESP32 board package is installed in the Arduino IDE to enable programming and uploading code to the ESP32 development board. It provides the required board definitions and compilation support.

4.USB Driver

The appropriate USB-to-serial driver allows the computer to communicate with the ESP32 through a USB connection. It helps the Arduino IDE detect the board and upload the program.

5.LCD Library

The required LCD library is used to display text and system status messages. The specific library depends on the type of LCD module and its connection method.

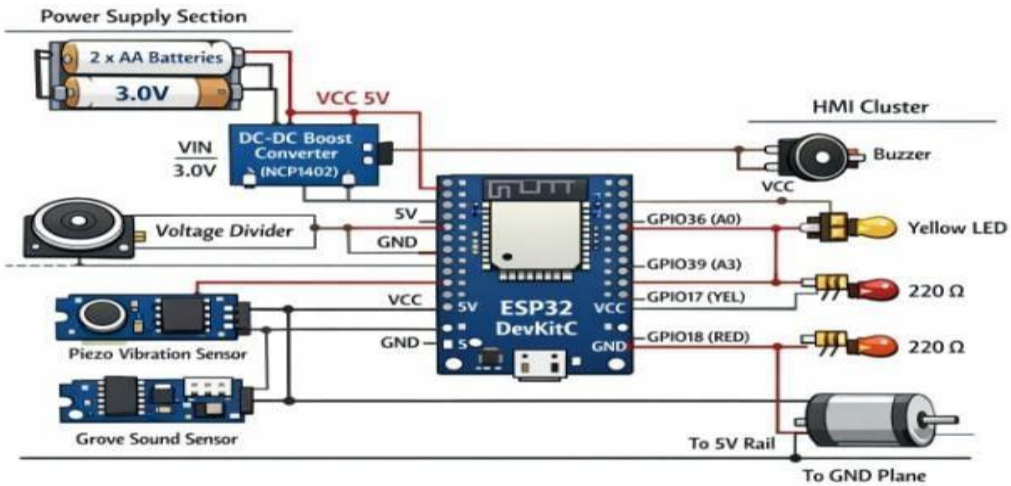
6.Serial Monitor

The Arduino IDE Serial Monitor is used to observe sensor readings and debugging messages during testing. It helps verify whether the sensor input is received and whether the programmed conditions work correctly.

These software tools support the development, programming, and testing of the Track Shield prototype. Together with the hardware components, they enable vibration monitoring and alert generation. Advanced AI analysis and remote IoT monitoring would require additional software development if incorporated into future versions.

CHAPTER 4. IMPLEMENTATION

4. IMPLEMENTATION



4.1 METHODOLOGY

The methodology of Track Shield describes the step-by-step process used to detect abnormal vibrations in a railway track model and generate alerts. The system uses an ESP32

microcontroller, a vibration sensor, an LCD display, and a buzzer. The hardware components are connected and programmed to work together for monitoring and alert generation.

Step 1: Hardware Setup

The ESP32 microcontroller, vibration sensor, LCD display, and buzzer are connected according to the circuit design. The vibration sensor is positioned on the railway track model to detect disturbances.

Step 2: Sensor Input

The vibration sensor detects vibrations or disturbances and sends the corresponding output signal to the ESP32. The sensor readings depend on the type of sensor and its configuration.

Step 3: Data Processing

The ESP32 receives the sensor output and processes it using the uploaded program. The program checks the sensor signal against a predefined condition or threshold to determine whether an abnormal vibration has been detected.

Step 4: Status Display

The LCD display shows the current monitoring status based on the sensor input. This allows the user to observe the system response during testing.

Step 5: Alert Generation

When the sensor output meets the programmed abnormal-vibration condition, the ESP32 activates the buzzer. The audible alert notifies the user that a disturbance has been detected.

Step 6: Testing and Evaluation

The prototype is tested under controlled conditions by introducing vibrations to the railway track model. The sensor response, LCD messages, and buzzer operation are observed to verify that the system works as programmed.

Step 7: Result Observation

The system’s performance is evaluated by checking whether the sensor detects vibrations and whether the LCD and buzzer respond correctly. The observations are used to identify possible improvements in sensor sensitivity and system reliability.

4.2 IMPLEMENTATION The implementation of Track Shield involves assembling the hardware components, uploading the program to the ESP32 microcontroller, and testing the system under controlled conditions. The system is designed to detect vibration disturbances and provide visual and audible alerts.

The implementation is divided into the following modules:

1. Vibration Detection Module: Detects vibrations and disturbances using the vibration sensor.

2. Data Processing Module: Processes the sensor output using the ESP32 and checks the programmed conditions.

3. Status Display Module: Displays the system status and monitoring messages on the LCD.

4. Alert Generation Module: Activates the buzzer when the specified abnormal-vibration condition is detected.

5. Testing and Evaluation Module: Tests the prototype and verifies the response of the sensor, LCD, and buzzer.

These modules work together to demonstrate basic railway track disturbance monitoring. The system's performance is evaluated by observing sensor responses, display messages, and buzzer alerts during testing. The current implementation focuses on vibration-based detection; actual rail crack identification and comprehensive intrusion detection require additional sensors and validation.

4.2.1 VIBRATION DETECTION MODULE

The vibration detection module is an important part of the Track Shield system. It uses a vibration sensor to detect vibrations and disturbances occurring in the railway track model. The sensor converts the detected vibration into an electrical output signal and sends it to the ESP32 microcontroller.

The ESP32 receives the sensor signal and processes it according to the programmed logic. When the sensor output satisfies the predefined abnormal-vibration condition, the system identifies the disturbance and initiates the corresponding response.

The vibration sensor is connected to the ESP32 using suitable input and power connections. During testing, vibrations are introduced to the track model to observe the sensor response. The sensor readings and system behaviour are checked to ensure that the module operates as expected.

This module provides the input required for the monitoring process. However, the vibration sensor detects disturbances rather than directly identifying the exact cause. Additional sensors and further testing would be necessary to distinguish actual railway track damage from other sources of vibration.

4.2.2 DATA PROCESSING MODULE

The data processing module is responsible for processing the signals received from the vibration sensor. The ESP32 microcontroller acts as the main processing unit of the Track Shield system. It receives the sensor output and evaluates it according to the conditions defined in the program.

When the sensor detects a vibration, the ESP32 processes the input and checks whether it meets the predefined threshold. Based on this condition, the microcontroller determines the appropriate system response. If an abnormal

vibration condition is detected, it updates the LCD display and activates the buzzer.

The processing module coordinates the operation of the connected hardware components and ensures that the system responds according to the programmed instructions. It also supports testing by allowing the sensor input and system responses to be observed.

This module forms the decision-making part of the prototype. The current implementation uses programmed conditions to identify abnormal vibrations. Advanced AI-based analysis could be incorporated in future versions to classify different vibration patterns after collecting and validating suitable data.

4.2.3 STATUS DISPLAY MODULE

The status display module is used to display the monitoring status of the Track Shield system. An LCD display is connected to the ESP32 microcontroller to show messages based on the input received from the vibration sensor.

The ESP32 processes the sensor output and updates the LCD according to the programmed conditions. During normal operation, the display indicates the system status. When an abnormal vibration condition is detected, the LCD displays a corresponding alert message to inform the user about the detected disturbance.

The LCD module helps users monitor the system without continuously checking the sensor output through a computer. It also makes the prototype easier to demonstrate and test by providing clear visual information.

This module improves the usability of Track Shield by displaying the system's response in real time, depending on the program and hardware configuration. The displayed message indicates a vibration condition and does not independently confirm actual railway track damage.

4.2.4 ALERT GENERATION MODULE

The alert generation module is responsible for notifying the user when an abnormal vibration condition is detected in the Track Shield system. It uses a buzzer connected to the ESP32 microcontroller to provide an audible warning.

When the vibration sensor detects a disturbance, the ESP32 processes the sensor signal and checks it against the predefined condition in the program. If the condition for abnormal vibration is satisfied, the ESP32 activates the buzzer. The sound produced by the buzzer alerts the user that a disturbance has been detected in the railway track model.

The buzzer works together with the vibration sensor, ESP32, and LCD display to provide a simple alert mechanism. The LCD displays the corresponding status, while the buzzer provides an audible indication. This helps the user identify the system's response during prototype testing.

The alert generation module improves the effectiveness of the Track Shield prototype by providing immediate notification when the programmed condition is met. However, the alert indicates an abnormal vibration and does not independently confirm a railway track crack or

intrusion. Further development and testing are necessary before using this system in real railway environments.

4.2.5 TESTING AND EVALUATION MODULE

The testing and evaluation module is used to verify the functionality and performance of the Track Shield prototype. The system is tested under controlled conditions to check whether the vibration sensor detects disturbances and whether the ESP32 responds correctly.

During testing, vibrations are introduced to the railway track model, and the sensor output is observed. The ESP32 processes the received signals according to the programmed conditions. The LCD display is checked to confirm that the appropriate status message appears, and the buzzer is tested to verify that it activates when the predefined abnormal-vibration condition is satisfied.

The following tests are performed:

1. Sensor Testing: Checks whether the vibration sensor responds to disturbances.

2. Processing Testing: Verifies whether the ESP32 processes sensor signals according to the program.

3. LCD Testing: Confirms that the display shows the appropriate system status.

4. Buzzer Testing: Checks whether the buzzer activates under the programmed alert condition.

5. System Integration Testing: Verifies that all connected components work together as expected.

The observations obtained during testing are used to identify connection errors, improve sensor sensitivity, and verify the operation of the prototype. The evaluation focuses on vibration detection and alert generation under controlled conditions.

The current prototype demonstrates basic vibration-based monitoring. Further experiments using different vibration levels, additional sensors, and validated detection techniques are required to evaluate its ability to identify specific railway track defects or intrusion events.

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

Track Shield is a railway track monitoring prototype designed to detect abnormal vibrations and provide alerts using sensor-based technology. The system uses an ESP32 microcontroller, a vibration sensor, an LCD display, and a buzzer to monitor disturbances in a railway track model.

The vibration sensor detects disturbances and sends signals to the ESP32, which processes the input according to the programmed conditions. The LCD displays the system status, while the buzzer provides an audible alert when an abnormal vibration condition is detected.

The project demonstrates how low-cost electronic components can be integrated to develop a basic railway track disturbance monitoring system. It also provides practical experience in microcontroller programming, sensor interfacing, and hardware integration.

Although the prototype demonstrates vibration detection and alert generation, it cannot independently confirm actual railway track cracks or identify every type of intrusion. Further improvements, additional sensors, and extensive testing are required to develop a reliable railway safety monitoring system. Overall, Track Shield provides a foundation for exploring intelligent and automated railway track monitoring technologies.

5.2FUTURE WORK

The Track Shield system can be improved in several ways to enhance its monitoring capabilities and reliability. Future developments may include the following:

- **AI-Based Detection:** Machine learning techniques can be incorporated to analyse vibration patterns and distinguish different types of disturbances after collecting and validating suitable data.

- **Additional Sensors:** Suitable sensors can be added to improve the detection of track defects and possible intrusion events.

- **IoT-Based Remote Monitoring:** IoT connectivity can be integrated to transmit monitoring information to a remote dashboard, allowing users to observe the system status from a distance.

- **Mobile Notifications:** Alert notifications can be developed to inform authorised users when abnormal conditions are detected.

- **Improved Accuracy:** The system can be tested under different conditions to improve sensor sensitivity, reduce false alerts, and increase reliability.

- **Real-Time Monitoring:** Continuous monitoring and data recording can be developed to help identify changes in vibration patterns over time.

- **Real-World Testing:** The improved system must undergo extensive testing and validation before it can be considered for use in actual railway environments.

These future enhancements can help develop Track Shield into a more advanced railway monitoring system. By combining validated AI techniques, suitable sensors, and IoT connectivity, the project may provide better support for railway track condition monitoring and safety.

5.3 SOURCE CODE

The Track Shield system is programmed using Embedded C/C++ in the Arduino IDE. The program reads the vibration sensor output, processes the input using the ESP32 microcontroller, displays the system status on the LCD, and activates the buzzer when the programmed abnormal-vibration condition is detected.

The source code is developed to integrate the sensor, microcontroller, display, and alert unit into a single monitoring prototype. The program can be modified according to the hardware connections and sensor configuration.

```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_ADXL345_U.h>
#include <LiquidCrystal_I2C.h>

// I2C pins for ESP32
#define SDA_PIN 21
#define SCL_PIN 22

// Sensor pins
#define VIBRATION_PIN 27
#define BUZZER_PIN 26

// LCD address and size
LiquidCrystal_I2C lcd(0x27, 16, 2);

// ADXL345 accelerometer
Adafruit_ADXL345_Unified accel = Adafruit_ADXL345_Unified(12345);

// Vibration threshold
float baseline = 9.8;
float threshold = 3.0;

void setup() {
```

```
Serial.begin(115200);

// Start I2C
Wire.begin(SDA_PIN, SCL_PIN);

pinMode(VIBRATION_PIN, INPUT);
pinMode(BUZZER_PIN, OUTPUT);

digitalWrite(BUZZER_PIN, LOW);

// LCD initialization - using your code
lcd.init();
lcd.backlight();
lcd.clear();

lcd.setCursor(0, 0);
lcd.print("TRACK SHIELD");
lcd.setCursor(0, 1);
lcd.print("Starting...");
delay(2000);

// Initialize ADXL345
if (!accel.begin()) {
  Serial.println("ADXL345 not detected!");
```

```
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ADXL345 ERROR");
  lcd.setCursor(0, 1);
  lcd.print("Check sensor");

  while (true) {
    digitalWrite(BUZZER_PIN, LOW);
    delay(1000);
  }
}

// Set accelerometer range
accel.setRange(ADXL345_RANGE_16_G);

// Measure baseline acceleration
sensors_event_t event;
accel.getEvent(&event);

baseline = sqrt(
  event.acceleration.x * event.acceleration.x +
  event.acceleration.y * event.acceleration.y +
  event.acceleration.z * event.acceleration.z
```

```
  );

  lcd.clear();
}

void loop() {
  // Read vibration sensor
  int vibration = digitalRead(VIBRATION_PIN);

  // Read ADXL345
  sensors_event_t event;
  accel.getEvent(&event);

  float magnitude = sqrt(
    event.acceleration.x * event.acceleration.x +
    event.acceleration.y * event.acceleration.y +
    event.acceleration.z * event.acceleration.z
  );

  float deviation = abs(magnitude - baseline);

  // Alert if either sensor detects vibration
  bool alert = (vibration == HIGH || deviation > threshold);
```

```
lcd.clear();

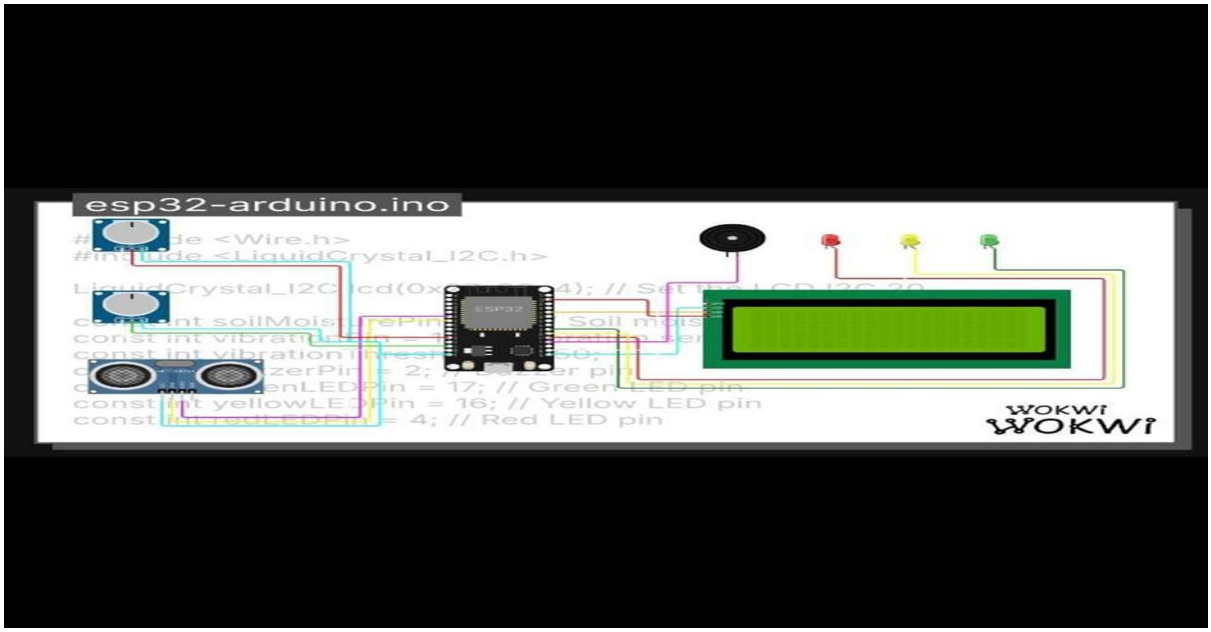
if (alert) {
  lcd.setCursor(0, 0);
  lcd.print("VIBRATION");
  lcd.setCursor(0, 1);
  lcd.print("DETECTED");

  digitalWrite(BUZZER_PIN, HIGH);
  Serial.println("VIBRATION DETECTED");
} else {
  lcd.setCursor(0, 0);
  lcd.print("VIBRATION");
  lcd.setCursor(0, 1);
  lcd.print("NOT DETECTED");

  digitalWrite(BUZZER_PIN, LOW);
  Serial.println("VIBRATION NOT DETECTED");
}

// Display accelerometer reading in Serial Monitor
Serial.print("Acceleration deviation: ");
Serial.print(deviation, 2);
Serial.println(" m/s^2");
```

```
  delay(500);
}
```



5.4 RESULTS

The Track Shield prototype is designed to monitor vibrations in a railway track model using an ESP32 microcontroller and a vibration sensor. The sensor provides input to the microcontroller, which processes the signal according to the programmed conditions.

The LCD display shows the system status, and the buzzer provides an audible alert when the specified abnormal-vibration condition is detected. The prototype demonstrates the integration of sensor input, microcontroller processing, visual output, and audible alerts in a single system.

The expected result is a basic railway track disturbance monitoring system that can respond to vibrations under controlled testing conditions. The performance of the prototype can be evaluated by observing the sensor response, LCD messages, and buzzer operation.

The system provides a foundation for further development using additional sensors, validated AI-based analysis, and IoT connectivity. Actual performance results should be recorded after testing the completed hardware. The current prototype does not independently confirm railway track cracks or detect all types of intrusion.

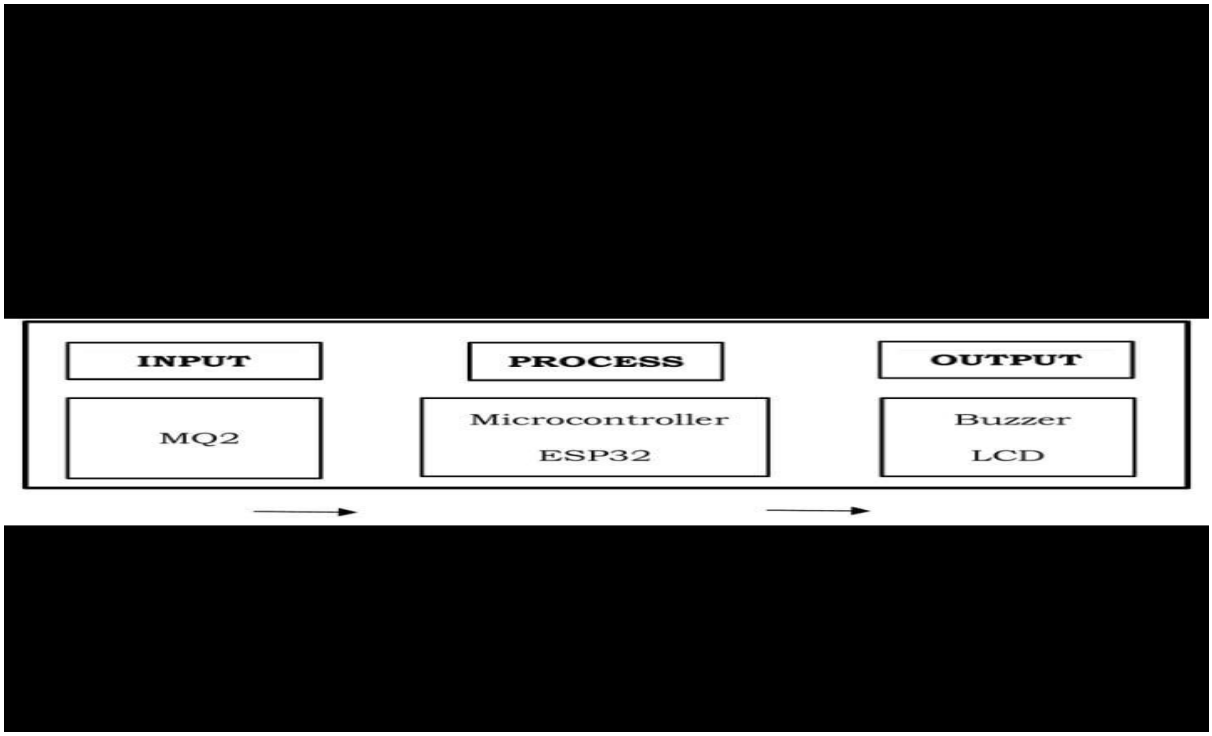
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Track shield: An intelligent AI IOT system for Railway track health and intrusion detection

held during 21.10.2026 & 22.10.2026 organized by the Institute of CSE, SIMATS Engineering, SIMATS, Thandalam, Chennai. Your presence and engagement have contributed significantly to the discussions and knowledge-sharing on AI-driven Engineering Innovations.

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